

Complete High-Frequency and Ultra-Tail Closure at the Original Physical Normalization:

An Exact Partition, an Optimal Hilbert Test,
and a Sharp Phase-Blind Obstruction

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Abstract

The TPC reconstruction program has several rigorous low-window interfaces, but its working Hypothesis H4 asks for more: every high-frequency, ultra-long, short-fiber, boundary, collision and polarization complement must return at the *original physical normalization*. A tail estimate in one row or one favorable sector does not by itself meet that quantifier.

We give a finite exact closure theorem conditional on a declared complete mask family. We characterize exact one-fold partition of the literal physical atom set into the low window and all complement classes. On a frequency class, unitary Fourier coordinates give the optimal estimate

$$|\mathcal{T}_E| \leq \nu_X \left(\sum_{r \in E} |\widehat{m}_X(r)|^2 \right)^{1/2} \left(\sum_{r \in E} |\widehat{A}_X(r)|^2 \right)^{1/2};$$

the constant one is sharp. Combining this with complete outer absolute-mass bounds for soft classes and literal aggregate packet bounds for retained classes yields a sufficient original-scale closure. Every reconstruction multiplier, provenance multiplicity and outer normalization must occur inside those returned bounds.

We also prove a sharp obstruction. If only the magnitudes of the physical atoms are known, the worst coherent choice of phases has size exactly their total absolute mass. Hence support sparsity, shortness of individual fibers, or a partition into many blocks cannot yield a power saving without coefficient-sensitive cancellation or an absolute-mass theorem. Existing TPC low-window tail reserves therefore remain valid local inputs, but H4 is not closed until their complete outer transport and every ultra-tail class are certified. The result is exact L0/L1 bookkeeping; it proves no new L2 fixed-shift estimate, parity breakthrough or twin-prime theorem.

Keywords: high-frequency complement; ultra-tail; physical normalization; Fourier energy; phase alignment; stop-go certificate.

Physical convention. One fixed nonzero shift h_0 , the literal active atom set, both polarizations, all reconstruction multipliers, and one inherited global normalization ν_X are retained. No class may be dropped because it is inconvenient or re-normalized after being estimated.

1 Why a complete tail certificate is still needed

TPC-93 exports the complete literal low-frequency window and, under its stated moment/truncation inequality, gives a fixed-power reserve for the associated high-frequency reconstruction tail [3]. It explicitly leaves the short, resonant and generic returned sectors to later arithmetic estimates. TPC-102 records that closing one resonant branch still leaves the high-frequency return, block-to-physical normalization and the strict endpoint ledger [2]. TPC-MVP1 packages these requirements as Hypothesis H4 [1].

The logical issue is a mismatch of quantifiers. A bound for one frequency row has the form

$$|\mathcal{T}_{E,\alpha}| \leq B_{E,\alpha}.$$

H4 asks for the scalar obtained only *after* all outer labels α , signs, masks and multipliers have been reassembled. Summing the displayed row bounds is valid, but its total cost may erase the saving. Keeping the signed sum may be much better, but then its cancellation must be proved. The definitions below make the two routes explicit.

2 The literal atom partition

For each X , let Ω_X be the finite set of inherited physical atoms. An atom ω contains every outer label, frequency, polarization and provenance key. Let

$$z_X(\omega) = c_X(\omega)A_X(\omega) \in \mathbb{C}$$

be its literal scalar contribution before the one global normalization $\nu_X > 0$. Thus

$$\mathcal{H}_X = \nu_X \sum_{\omega \in \Omega_X} z_X(\omega).$$

Definition 2.1 (Complete complement partition). A family of masks

$$\mathbf{1}_{\text{Low}_X}, \quad \mathbf{1}_{\text{High}_X}, \quad \mathbf{1}_{\text{Ultra}_X}, \quad \mathbf{1}_{\text{Short}_X}, \quad \mathbf{1}_{\text{Boundary}_X}, \quad \mathbf{1}_{\text{Collision}_X}$$

is complete if each mask is 0-1, their supports are pairwise disjoint, and their sum is $\mathbf{1}_{\Omega_X}$. Empty classes are allowed. Further literal classes may be added, but no atom may have coverage zero or greater than one.

Proposition 2.2 (Exact original-scale disintegration). *For a complete partition $\mathcal{C}_X$,*

$$\mathcal{H}_X = \sum_{C \in \mathcal{C}_X} \mathcal{H}_{X,C}, \quad \mathcal{H}_{X,C} = \nu_X \sum_{\omega \in C} z_X(\omega).$$

If the masks overlap with multiplicity $d_X(\omega)$, the displayed sum instead equals $\mathcal{H}_X + \nu_X \sum_{\omega} (d_X(\omega) - 1)z_X(\omega)$. If they miss atoms, the corresponding omitted sum remains as a literal residual.

Proof. Insert $\sum_C \mathbf{1}_C(\omega) = 1$ into the finite sum. The overlap and missing formulas follow by retaining the actual coverage multiplicity. $\square$

Remark 2.3 (Classification is not estimation). [Proposition 2.2](#) is an exact L0 identity. It does not say that any class is small. In particular, “ultra-long” and “short” are geometric names, not analytic bounds.

3 An optimal frequency-tail test

Let G be a finite abelian frequency group and use a unitary Fourier transform. A literal row has the form

$$\mathcal{R}_X = \nu_X \sum_{r \in G} \hat{m}_X(r) \overline{\hat{A}_X(r)}.$$

The multiplier $\hat{m}_X$ includes the signed reconstruction filter. The response $\hat{A}_X$ includes the complete coefficient and provenance aggregation for that row.

Theorem 3.1 (Sharp Hilbert tail certificate). *For every frequency set $E \subseteq G$,*

$$|\mathcal{T}_E| := \nu_X \left| \sum_{r \in E} \widehat{m}_X(r) \overline{\widehat{A}_X(r)} \right| \leq \nu_X M_E^{1/2} A_E^{1/2},$$

where

$$M_E = \sum_{r \in E} |\widehat{m}_X(r)|^2, \quad A_E = \sum_{r \in E} |\widehat{A}_X(r)|^2.$$

The constant 1 is optimal. If $\widehat{m}_X \mathbf{1}_E \neq 0$, equality is attained by taking $\widehat{A}_X \mathbf{1}_E = t \widehat{m}_X \mathbf{1}_E$ for any $t \geq 0$.

Proof. This is Cauchy–Schwarz in $\ell^2(E)$. The stated proportional choice makes the two vectors collinear and gives equality. $\square$

Corollary 3.2 (Callable high-frequency return). *A proved pair of complete outer estimates*

$$M_{\text{High}_X} \leq X^{\mu+o(1)}, \quad A_{\text{High}_X} \leq X^{\alpha+o(1)}$$

returns

$$|\mathcal{T}_{\text{High}_X}| \leq \nu_X X^{(\mu+\alpha)/2+o(1)}.$$

This Hilbert certificate yields $o(X)$ exactly when its displayed right-hand side, with the literal ν_X , is $o(X)$. The actual tail can still be smaller through cancellation not used by the certificate. A rowwise version is callable only after the sum of its rowwise right-hand sides is shown to have that same property, or after a coherent outer Hilbert estimate is proved.

Remark 3.3 (What the theorem does not manufacture). Parseval may identify A_E with a physical energy, but it does not make that energy small. Nor does the label “high frequency” force M_E to decay. The two quantitative inputs in [corollary 3.2](#) must come from the literal filter and arithmetic response.

4 Sharp phase-blind obstruction

The alternative to an energy estimate is often to bound every atom or block separately. The next elementary identity is the exact limit of that strategy.

Theorem 4.1 (Worst coherent completion). *Let $a_\omega \geq 0$ be prescribed magnitudes on a finite set Ω . Then*

$$\sup_{|\varepsilon_\omega|=1} \left| \sum_{\omega \in \Omega} \varepsilon_\omega a_\omega \right| = \sum_{\omega \in \Omega} a_\omega.$$

Consequently, from magnitude data $|z_X(\omega)| \leq a_\omega$ alone, the best uniform original-scale bound is

$$|\mathcal{H}_X| \leq \nu_X \sum_{\omega} a_\omega,$$

and the constant one is sharp.

Proof. The triangle inequality gives the upper bound. Taking every $\varepsilon_\omega = 1$ attains it. $\square$

Corollary 4.2 (No support-only tail saving). *Knowing only that a complement is sparse, decomposes into short fibers, or has bounded overlap cannot give an $o(X)$ bound unless those facts imply*

$$\nu_X \sum_{\omega \in \text{complement}} |z_X(\omega)| = o(X).$$

Otherwise a coefficient-sensitive signed theorem, an energy theorem with quantitative input, or a different exact representation is necessary.

Remark 4.3 (A sharp local obstruction is not a global lower bound). [Theorem 4.1](#) says that the *information supplied* cannot exclude alignment. It does not assert that the actual arithmetic coefficients align, and therefore proves no lower bound for the physical tail. Its conclusion is a stop certificate for a phase-blind proof route, not for the prime-pair problem.

5 Complete conditional closure

Split the complement classes into soft classes S_X and retained packet classes R_X . “Soft” means that an original-scale absolute or coherent estimate is actually available; it is not a geometric synonym.

Theorem 5.1 (Original-scale complement closure and H4 specialization). *Assume:*

- (i) *the masks form a complete partition in the sense of [definition 2.1](#);*
- (ii) *the complete signed low window is retained without changing its normalization;*
- (iii) *the soft complements satisfy*

$$\sum_{C \in S_X} |\mathcal{H}_{X,C}| = o(X);$$

- (iv) *after all reconstruction, provenance and outer reassembly costs, the retained complement satisfies the aggregate estimate*

$$\left| \sum_{C \in R_X} \mathcal{H}_{X,C} \right| \leq X^{1-\sigma+o(1)} \quad \text{for some fixed } \sigma > 0.$$

Then the complete complement is $o(X)$, and

$$\mathcal{H}_X = \mathcal{H}_{X, \text{Low}_X} + o(X)$$

at the original physical normalization. A high-frequency class may meet item (iii) or (iv) through [theorem 3.1](#), but only with its complete outer energies. In addition, item (iv) supplies the retained-packet clause of MVP1 Hypothesis H4 only when one may take

$$\sigma \geq \sigma_* = \frac{1}{400}.$$

An arbitrary fixed $\sigma > 0$ proves the displayed $o(X)$ complement closure, but does not by itself preserve the raw H4 packet reserve. Any later physical reassembly loss is charged separately in H9.

Proof. Apply [proposition 2.2](#), then the hypotheses to the two disjoint complement sums. Since a fixed power saving is $o(X)$, their total is $o(X)$. $\square$

Remark 5.2 (Per-packet estimates are insufficient by themselves). If there are N_X retained packets and only $|\mathcal{H}_{X,C}| \leq X^{1-\sigma+o(1)}$ is known separately, triangle summation costs N_X . The theorem requires the resulting aggregate exponent, including this cost, not the best packet exponent quoted in isolation.

6 Finite exact regression

The standard-library script `experiments/tpc116_tail_certificate.py` uses rational arithmetic to verify:

1. exact one-fold coverage and exact atom disintegration;
2. the explicit overlap-defect identity;
3. Cauchy–Schwarz for rational tail vectors and equality on proportional vectors;
4. the exact phase-blind supremum for all real sign assignments in finite examples; and
5. a soft/retained closure ledger that refuses incomplete masks and unknown class bounds.

The committed JSON is an algebra regression only. It contains no computed prime-pair evidence and no asymptotic high-tail estimate.

7 Verdict and next gate

Status	Required literal certificate	Consequence
Go for $o(X)$ closure	Complete masks and complete outer soft/retained bounds satisfying theorem 5.1	The complement is $o(X)$ at the original normalization; pass the same atoms to the TPC-18 cover audit.
Go for the H4 packet interface	The preceding certificate with retained exponent $\sigma \geq \sigma_* = 1/400$	The raw H4 reserve is preserved; H9 re-assembly costs remain separate.
Stop phase-blind route	Only support/magnitude data are known and their total absolute mass is not $o(X)$	Theorem 4.1 is sharp; retain signs or prove energy cancellation.
Stop declaration	An atom is missing, duplicated, or returned at a changed normalization	The claimed complete complement is not the original coefficient.
Current TPC state	Local high-tail reserves exist, but no archived complete outer certificate for every H4 class is supplied here	H4 remains open; no zero loss may be entered in the endpoint ledger.

Established. The partition identity, optimal Hilbert bound, phase-blind supremum, conditional complement closure and H4 specialization are exact L0 statements. Their attachment to the literal TPC labels and normalization is an L1 interface.

Not established. We do not prove the complete outer energy or absolute mass of any growing TPC high/ultra-tail carrier. We do not verify every H4 atom from a machine-readable coefficient archive. No new L2 fixed- h_0 saving, parity breaking, Hardy–Littlewood asymptotic, prime-pair lower bound or twin-prime theorem is obtained. The next gate is a literal TPC-18 block cover and physical residual audit.

References

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